EXPERIMENT 6 – INPUT AND OUTPUT

1. User-defined package – Prime number and factorial

Input

```
from my_package import module1 as m

number = int(input("\nEnter number = "))
print(m.check_prime(number))
print(m.calculate_factorial(number))
```

Output

```
Enter number = [Enter a number]

The uploaded source file does not show the program output.
```

2. User-defined package – Addition and multiplication

Input

```
from my_package import module2 as m

a, b = map(int, input("Enter two numbers separated by a space: ").split())
print(m.add(a, b))
print(m.mul(a, b))
```

Output

```
Enter two numbers separated by a space: [Enter two numbers]

The uploaded source file does not show the program output.
```

3. Math package – Square root

Input

```
import math
print(math.sqrt(16))

from math import sqrt
print(sqrt(25))
```

Output

```
4.0
5.0
```

4. NumPy – Create an array using arange()

Input

```
import numpy as np
a = np.arange(6)
a
```

Output

```
array([0, 1, 2, 3, 4, 5])
```

5. NumPy – One-dimensional array

Input

```
a = np.array([2, 3, 4])
a
```

Output

```
array([2, 3, 4])
```

6. NumPy – Two-dimensional array

Input

```
b = np.array([(1.5, 2, 3), (4, 5, 6)])
print(b)
```

Output

```
[[1.5 2.  3. ]
 [4.  5.  6. ]]
```

7. NumPy – Complex data type

Input

```
c = np.array([(1.5, 2, 3), (4, 5, 6)], dtype=complex)
c
```

Output

```
array([[1.5+0.j, 2. +0.j, 3. +0.j],
       [4. +0.j, 5. +0.j, 6. +0.j]])
```

8. NumPy – Reshape array

Input

```
a = np.arange(15).reshape(3, 5)
a
```

Output

```
array([[ 0,  1,  2,  3,  4],
       [ 5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14]])
```

9. NumPy – Array subtraction

Input

```
a = np.array([20, 30, 40, 50])
print(a)
b = np.arange(4)
print(b)
c = a - b
print(c)
```

Output

```
[20 30 40 50]
[0 1 2 3]
[20 29 38 47]
```

10. NumPy – Slicing rows and alternate columns

Input

```
a = np.array([[-1, 2, 0, 4],
              [4, -0.5, 6, 0],
              [2.6, 0, 7, 8],
              [3, -7, 4, 2.0]])

a = a[:2, ::2]
print("first 2 rows and alternate columns(0 and 2):\n", a)
```

Output

```
first 2 rows and alternate columns(0 and 2):
[[-1.  0.]
 [ 4.  6.]]
```